
What is JDSS?

The Joint Dismounted Soldier System - STANAG 4677 / AEP-76

JDSSArrow - a SOLID, pluggable reference implementation of the Joint Dismounted Soldier System (NATO STANAG 4677, specified in AEP-76, 5 volumes), with a web configuration and monitoring application.

Reference / educational software implementing a published NATO interoperability standard. Not accredited or production tactical software.

1. Overview

The Joint Dismounted Soldier System (JDSS) is a NATO interoperability standard - STANAG 4677, technically specified in Allied Engineering Publication AEP-76 (Edition A, 5 volumes). It provides soldier-level Command, Control, Communications and Computers (C4) interoperability at the tactical edge: the layer below HQ-level Friendly Force Tracking (NFFI / STANAG 5527).

National soldier systems - each built by a different nation with different radios, data formats and security - connect to a shared coalition network through a JDSS Gateway. The gateway composes the five AEP-76 volumes so that dismounted troops from different nations can exchange Command and Control (C2) data directly.

The loaned radio concept

The central idea of JDSS is the loaned radio: a coalition partner is given (loaned) a radio bearer that plugs their national soldier system into the coalition network. This enables direct C2 data exchange between coalition dismounted troops without every nation having to adopt the same national system - they only need to speak JDSS at the gateway.

Where JDSS sits

- HQ / staff level: Friendly Force Tracking via NFFI (STANAG 5527) and JC3IEDM data.
- Soldier / tactical edge: JDSS (STANAG 4677 / AEP-76) - the focus of this standard.
- JDSS is deliberately the layer below FFT: it is about the individual dismounted soldier and small-team situational awareness, not the recognised air/maritime picture.

2. The five AEP-76 volumes

AEP-76 is organised into five volumes, each covering one concern of interoperability. A JDSS Gateway is precisely a composition of one implementation of each volume.

Vol	AEP-76 subject			What it defines
I	Security			Authenticity and integrity of every frame; classification marking and releasability caveats.
II	Data Model (JDSSDM)			The message content model - a MIP-3.1-XML variant, JC3IEDM-compliant, using APP-6(D) symbology.
III	Loaned Radio			The radio bearer abstraction that carries JDSS traffic over a tactical link.
IV	Information	Exchange	Mechanism	How messages are framed, transported and reliably delivered between nodes.
V	Network Access			How a node obtains its addressing on the coalition network (e.g. deriving the multicast group).

3. The JDSS Data Model (JDSSDM)

The JDSSDM (Volume II) is the shared language of the network. It is a MIP-3.1-XML variant, JC3IEDM-compliant, and uses APP-6(D) military symbology (Symbol Identification Codes) so that a contact reported by one nation renders as the correct symbol on every other nation's map.

Every JDSS message carries a header (originator, message id, reporting time, classification, releasability) and a typed body. There are seven message types:

Message type	Purpose
Presence	Periodic position/status beacon ('I am here') - drives the blue-force picture and peer liveness.
Identification	Who a node is: callsign, unit, nation, role.
ContactSighting	A report of an observed contact/target, with location and APP-6(D) symbol.
Sketch	A free-form sketch (ordered points) - e.g. a route or a hasty diagram.
Overlay	Tactical control measures / map graphics (boundaries, markers).
CasevacRequest	A casualty evacuation request (the 9-line), with urgency.
Chat	Free-text tactical chat, optionally addressed to a recipient.

4. Security and classification

Volume I makes every frame authenticated and integrity-protected. In this reference implementation the default security provider signs each payload with HMAC-SHA256 keyed by a coalition pre-shared key (PSK). A frame whose HMAC does not verify is dropped before it can reach the operational picture - this is how an unauthorised (wrong-key) node is kept off the network.

Every originated message is stamped with a classification level and a releasability caveat:

Level	Marking	Releasable-to example
0	UNCLASSIFIED	ALL
1	RESTRICTED	REL BEL NLD
2	CONFIDENTIAL	REL <nations>
3	SECRET	REL <nations>

The receiving picture always shows the highest classification seen on the network, so an operator is never under-warned about the sensitivity of the data on their screen.

5. Glossary

Term	Meaning
JDSS	Joint Dismounted Soldier System (STANAG 4677).
AEP-76	The Allied Engineering Publication specifying JDSS in 5 volumes.
JDSSDM	JDSS Data Model (Volume II) - the message content model.
JDSSIEM	JDSS Information Exchange Mechanism (Volume IV).
Gateway	The composition of the five volumes that joins a node to the network.
Loaned radio	A radio bearer loaned to a partner to join the coalition network.
PSK	Pre-shared key - the coalition secret keying HMAC authentication.
APP-6(D)	NATO Joint Military Symbology (the symbol set used by JDSSDM).
JC3IEDM	Joint C3 Information Exchange Data Model that JDSSDM aligns with.
COP	Common Operational Picture - the shared tactical picture.